UNIVERSITY OF ILLINOIS AT URBANA-CHAMPAIGN
Department of Electrical and Computer Engineering

ECE 310 Digital Signal Processing

Homework 8

Profs. Kamalabadi, Snyder **Due: Friday, Oct 23, 11:59pm on Gradescope**

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1. The sequence $x[n] = \cos\left(\frac{\pi}{3}n\right),\ -\infty < n < \infty$ was obtained by sampling the continuous-time signal $x_a(t) = \cos(\Omega_0 t),\ -\infty < t < \infty$ at a sampling rate of 1000 samples/sec. What are two possible values of Ω_0 that could have resulted in the sequence $x[n]$?

2. The continuous-time signal $x_a(t) = \sin(10\pi t) + \cos(20\pi t)$ is sampled with a sampling period T to obtain the discrete-time signal $x[n] = \sin\left(\frac{\pi}{5}n\right) + \cos\left(\frac{2\pi}{5}n\right)$

   a) Determine a choice for T consistent with this information.

   b) Is your choice for T in part (a) unique? If so, explain why. If not, specify another choice of T consistent with the information given.

3. The continuous-time signal $x_a(t) = \cos(400\pi t)$ is sampled with a sampling period T to obtain a discrete-time signal $x[n] = x_a(nT)$

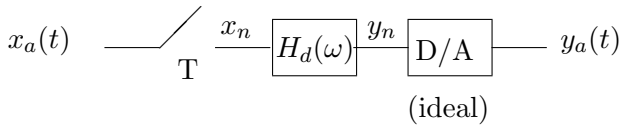
   a) Compute and sketch the magnitude of the continuous-time Fourier transform of $x_a(t)$ and the discrete-time Fourier Transform of $x[n]$ for $T = 1$ ms.

   b) Repeat part (a) for $T = 2$ ms.

   c) What is the maximum sampling period T_{max} such that no aliasing occurs in the sampling process?

4. The input to an D/A converter is $\{x[n]\} = \{-1, \underset{\uparrow}{0}, 0, 3\}$ with sampling interval T. Determine the output of the D/A converter if the D/A converter is (a) an ZOH, and (b) an ideal D/A.

5. Consider the following system with uniform sampling



   The discrete-time system $H_d(\omega)$ is an ideal low-pass filter with cutoff frequency $\frac{\pi}{8}$.

   (a) If $x_a(t)$ is bandlimited to 5 kHz, what is the maximum value of T that will avoid aliasing in the A/D converter?

   (b) If $\frac{1}{T} = 10$ kHz, determine the maximum bandwidth of $x_a(t)$ allowed such that the overall system from $x_a(t)$ to $y_a(t)$ behaves as an LTI system?

6. A speech signal $x_a(t)$ is assumed to be bandlimited to 12kHz. It is desired to filter this signal with a bandpass filter that will pass the frequencies between 300Hz and 6kHz by using a digital filter $H_d(\omega)$ sandwiched between an A/D and an ideal D/A.

   (a) Determine the Nyquist sampling rate for the input signal.

   (b) Sketch the frequency response $H_{d,1}(\omega)$ for the necessary discrete-time filter, when sampling at the Nyquist rate.

   (c) Find the largest sampling period T for which the A/D, digital filter response ($H_{d,2}(\omega)$), and D/A can perform the desired filtering function. (Hint: some amount of aliasing may be permissible during A/D conversion for this part.)

   (d) For the system using T from part (c), sketch the necessary $H_{d,2}(\omega)$.